The UMD-to-RMF Problem:
Audit of a Stability Argument and Further Partial Results

Prepared from the supplied source paper and partial solution packet

June 19, 2026

Abstract

Hytönen–McIntosh–Portal asked whether every UMD Banach space has the Rademacher maximal function property (RMF). I found no reliable proof or counterexample in the literature through June 19, 2026. The supplied partial result is nevertheless essentially correct: RMF passes to closed subspaces, quotients, and complex interpolation spaces. Two repairs are needed in the written proof: the quotient argument should include a Bochner lifting lemma, and the interpolation argument should use a one-sided interpolation embedding for finite Rademacher spaces rather than assert an unnecessary equality.

This note gives complete proofs of those closure results. It also proves that RMF is stable under Bochner L^q formation and finite representability, gives a finite-dimensional quantitative reformulation of the UMD-to-RMF problem, and derives an additional sufficient criterion—apparently not stated in the RMF references located—by combining a deterministic variation-to-R-bound estimate with the Pisier–Xu martingale variation theorem. These advances do not settle the general UMD question.

1 Executive conclusion

The question in the source paper is:

> Does every UMD Banach space have the RMF property?

The paper explicitly says that this was open and that a positive answer would remove the extra RMF assumption from its Banach-valued Kato theorem [3, Definition 2.12 and the paragraph following it].

My conclusions are as follows.

Claim in the supplied packet	Verdict
Closed subspaces preserve RMF	Correct, and explicitly recorded in the literature.
Quotients preserve RMF	Correct. The packet omits a proof that the induced map on Bochner L^p spaces is a metric quotient.
Complex interpolation preserves RMF	Correct after a small repair. A uniform one-sided embedding of finite Rademacher interpolation spaces is enough; a full interpolation identity need not be asserted.
Every UMD space has RMF	Not proved by the packet, and no reliable proof or counterexample was found in the literature search.

The strongest outcome of the present investigation is therefore a corrected closure theorem plus several additional partial results, not a complete resolution.

2 Literature status

The original examples with RMF are UMD function lattices, noncommutative L^q spaces for $1 < q < \infty$, and spaces of Rademacher type 2 [3, Proposition 2.13]. The same paper proves that ℓ^1 fails RMF. Kemppainen subsequently showed that RMF is independent of the standard choice of filtration and measure space, that weak and strong forms are equivalent, and that RMF forces nontrivial type; he also explicitly recorded inheritance by closed subspaces [5]. Hytönen–Kemppainen related RMF to vector-valued Carleson embeddings [4].

Later sources still describe the implication UMD $\Rightarrow$ RMF as open. This is stated explicitly, for example, by Amenta–Uraltsev in 2020 [1] and by Lorist in 2021 [6]. Later work sometimes avoids RMF assumptions in particular harmonic-analytic applications, but this does not prove the Banach-space implication itself; see, for example, [2].

There is one potentially misleading recent item. A 2025 textbook exercise asks the reader to prove that a UMD Banach space has RMF and gives the hint to use Rubio de Francia's extension of the Fefferman–Stein theorem [8, Chapter 5, exercise quoted in the indexed preview]. The standard Rubio de Francia argument proves RMF for *UMD Banach lattices*, where a lattice maximal function is available; it does not apply to arbitrary UMD spaces. In view of the explicit open-problem statements above and the lack of a supplied proof overcoming the non-lattice obstruction, the most plausible reading is that the word "lattice" was omitted. Absent an argument beyond the lattice case, the exercise is not treated here as a resolution.

I found no refereed article or arXiv preprint through June 19, 2026 that gives a full proof or a UMD counterexample.

3 Definitions and finite linearization

Let $(\Omega, \mathcal{F}, \mu)$ be a sigma-finite measure space with a filtration $(\mathcal{F}_j)_{j\in\mathbb{Z}}$, and let E_j denote conditional expectation onto $\mathcal{F}_j$. For a finite set $J \subset \mathbb{Z}$ and $f \in L^p(\Omega; X)$ define

$$M^X_{R,J} f(\omega) = \sup_{\|\lambda\|_{\ell^2(J)}\le 1} \mathbb{E}_\varepsilon \left\| \sum_{j\in J} \varepsilon_j \lambda_j E_j f(\omega) \right\|_X . \tag{1}$$

The space X has RMF$_p$ if

$$\|M^X_{R,J} f\|_{L^p(\Omega)} \le C \|f\|_{L^p(\Omega;X)} \tag{2}$$

uniformly in finite J. It is enough to verify this for one $p \in (1, \infty)$, and the property is independent of the standard filtration model [3, 5].

For interpolation it is convenient to use

$$\mathrm{Rad}^2_J(X) = \left\{ \sum_{j\in J} \varepsilon_j x_j : x_j \in X \right\} \subset L^2(\Omega_\varepsilon; X)$$

with the inherited L^2 norm. Kahane–Khintchine shows, with constants independent of J and X, that replacing the expectation in (1) by its $L^2(\Omega_\varepsilon)$ version gives an equivalent maximal function.

For finite J define the linear operator

$$T^X_J f(\omega)\lambda = \sum_{j\in J} \varepsilon_j \lambda_j E_j f(\omega), \qquad \lambda \in \ell^2(J). \tag{3}$$

Then the L^2 version of $M^X_{R,J} f(\omega)$ is exactly

$$\|T^X_J f(\omega)\|_{\mathcal{L}(\ell^2(J), \mathrm{Rad}^2_J(X))}.$$

4 Audit and corrected stability theorem

4.1 A Bochner quotient lemma

The quotient proof in the supplied packet uses a standard fact that merits an explicit proof.

Lemma 4.1 (Almost isometric Bochner lifting)**.** *Let $Q : X \to Z$ be a metric quotient map, so*

$$\|z\|_Z = \inf\{\|x\|_X : Qx = z\}.$$

For $1 \le p < \infty$, the induced map

$$\widetilde{Q} : L^p(\Omega; X) \longrightarrow L^p(\Omega; Z), \qquad \widetilde{Q}g = Qg,$$

is a metric quotient. Equivalently, for every $f \in L^p(\Omega; Z)$ and $\eta > 0$ there is $g \in L^p(\Omega; X)$ such that

$$Qg = f, \qquad \|g\|_{L^p(\Omega;X)} \le \|f\|_{L^p(\Omega;Z)} + \eta.$$

Proof. First suppose $h = \sum_{m=1}^{M} \mathbf{1}_{A_m} z_m$ is simple, with the A_m disjoint. Choose $x_m \in X$ with $Qx_m = z_m$ and $\|x_m\| \le (1+\delta)\|z_m\|$. Then $u = \sum_m \mathbf{1}_{A_m} x_m$ satisfies $Qu = h$ and $\|u\|_p \le (1+\delta)\|h\|_p$.

For general f, choose simple functions s_n such that $s_n \to f$ in L^p and so rapidly that, with $s_0 = 0$,

$$\sum_{n\ge1} \|s_n - s_{n-1}\|_p \le \|f\|_p + \eta/2.$$

This is obtained, for example, by requiring $\|f - s_n\|_p$ to decrease geometrically with a sufficiently small total sum. Set $h_n = s_n - s_{n-1}$. By the simple-function case choose u_n with $Qu_n = h_n$ and

$$\|u_n\|_p \le \|h_n\|_p + \eta 2^{-n-1}.$$

The series $\sum_n u_n$ converges absolutely in $L^p(\Omega; X)$ to a function g, and

$$\|g\|_p \le \sum_n \|u_n\|_p \le \|f\|_p + \eta.$$

Continuity of Q gives $Qg = \lim_n s_n = f$. □

4.2 The closure theorem

Theorem 4.2 (Subspaces, quotients, and complex interpolation)**.** *Fix $1 < p < \infty$.*

(i) *If X has* RMF_p *and $Y \subset X$ is a closed subspace, then Y has* RMF_p*.*

(ii) *If X has* RMF_p *and $Q : X \to Z$ is a quotient map, then Z has* RMF_p*.*

(iii) *If (X_0, X_1) is a compatible complex interpolation couple and both X_0 and X_1 have* RMF_p*, then $X_\theta = [X_0, X_1]_\theta$ has* RMF_p *for every $0 < \theta < 1$.*

The estimates are uniform over all finite truncations J.

Proof. For (i), if f is Y-valued, every $E_j f$ is Y-valued and the norm is the one inherited from X. Hence pointwise

$$M^Y_{R,J} f = M^X_{R,J} f.$$

For (ii), after an equivalent renorming of Z we may assume that Q is a metric quotient. Let $f \in L^p(\Omega; Z)$ and use Lemma 4.1 to choose $g \in L^p(\Omega; X)$ with $Qg = f$ and $\|g\|_p \le \|f\|_p + \eta$. Conditional expectation commutes with bounded linear maps, so $E_j f = QE_j g$. Since Q is contractive,

$$\mathbb{E}_\varepsilon \left\| \sum_{j \in J} \varepsilon_j \lambda_j E_j f(\omega) \right\|_Z \le \mathbb{E}_\varepsilon \left\| \sum_{j \in J} \varepsilon_j \lambda_j E_j g(\omega) \right\|_X .$$

Thus $M^Z_{R,J} f \le M^X_{R,J} g$. Take L^p norms and let $\eta \downarrow 0$.

For (iii), use the L^2 Rademacher norm described above. Put

$$A_i(J) = \mathcal{L}(\ell^2(J), \mathrm{Rad}^2_J(X_i)), \qquad i = 0, 1.$$

The endpoint assumptions say that the common operator T_J in (3) maps

$$L^p(\Omega; X_i) \longrightarrow L^p(\Omega; A_i(J))$$

with norm at most C_i, uniformly in J.

We use two contractive embeddings. First, evaluation on the common domain $\ell^2(J)$ gives

$$[A_0(J), A_1(J)]_\theta \hookrightarrow \mathcal{L}\left(\ell^2(J), [\mathrm{Rad}^2_J(X_0), \mathrm{Rad}^2_J(X_1)]_\theta\right). \tag{4}$$

Indeed, for each $\lambda \in \ell^2(J)$, the evaluation map $S \mapsto S\lambda$ has endpoint norm at most $\|\lambda\|_2$, and interpolation gives the claim.

Second,

$$[\mathrm{Rad}^2_J(X_0), \mathrm{Rad}^2_J(X_1)]_\theta \hookrightarrow \mathrm{Rad}^2_J([X_0, X_1]_\theta) \tag{5}$$

contractively, uniformly in J. To see this, regard each endpoint Rademacher space as a subspace of $L^2(\Omega_\varepsilon; X_i)$. Exactness of complex interpolation and the Bochner identity give a contractive map into

$$[L^2(\Omega_\varepsilon; X_0), L^2(\Omega_\varepsilon; X_1)]_\theta = L^2(\Omega_\varepsilon; [X_0, X_1]_\theta).$$

Every element F of the interpolation space already lies in the algebraic sum of the two finite Rademacher spans, hence has the form $F = \sum_{j \in J} \varepsilon_j x_j$ with $x_j \in X_0 + X_1$. The displayed inclusion places F in $L^2(\Omega_\varepsilon; [X_0, X_1]_\theta)$, and then $x_j = \mathbb{E}_\varepsilon(\varepsilon_j F) \in [X_0, X_1]_\theta$. This proves (5). Notice that no reverse inclusion and no uniformly bounded Rademacher projection are needed.

Finally, interpolate T_J and use

$$[L^p(\Omega; X_0), L^p(\Omega; X_1)]_\theta = L^p(\Omega; [X_0, X_1]_\theta)$$

and the analogous identity for the target Bochner spaces, followed by (4) and (5). We obtain

$$\|M^{X_\theta}_{R,J} f\|_p \lesssim_p C_0^{1-\theta} C_1^\theta \|f\|_{L^p(\Omega; X_\theta)},$$

where the only harmless factor is the uniform Kahane–Khintchine equivalence between the L^1 and L^2 Rademacher norms. □

Remark 4.3. *For an increasing sequence of finite sets J_n exhausting the filtration indices, $M_{R,J_n} f$ increases pointwise to the full Rademacher maximal function. Monotone convergence therefore passes the uniform finite estimates to the usual RMF estimate.*

Corollary 4.4 (Dual hypotheses in the Kato theorem)**.** *If both X and X^* have RMF and $Y \subset X$ is closed, then Y, Y^*, X/Y, and $(X/Y)^*$ all have RMF. If both endpoint spaces and their duals have RMF, then, under the usual duality hypotheses for a complex interpolation couple, the same holds for the interpolation space and its dual.*

Proof. For a closed subspace $Y \subset X$, one has $Y^* \cong X^*/Y^\perp$, and for a quotient $Z = X/Y$, one has $Z^* \cong Y^\perp \subset X^*$. Apply Theorem 4.2 to X and X^*. The interpolation assertion follows from the standard duality formula for regular complex interpolation couples. □

5 Additional closure results

For fixed $1 < p < \infty$, write $\rho_p(X)$ for the best RMF_p constant of X.

5.1 Bochner L^q spaces

Theorem 5.1 (Bochner stability)**.** *Let X have RMF and let (S, Σ, ν) be sigma-finite. For every $1 < q < \infty$, the Bochner space $L^q(S; X)$ has RMF. The same conclusion holds for $\ell^q(X)$ and for iterated Bochner spaces.*

Proof. By exponent independence of RMF, it suffices to prove the estimate with outer exponent q. Let $F \in L^q(\Omega; L^q(S; X))$. For fixed ω and $\lambda \in B_{\ell^2(J)}$, Kahane–Khintchine and Fubini give

$$\begin{aligned}
&\mathbb{E}_\varepsilon \left\| \sum_{j\in J} \varepsilon_j \lambda_j E_j F(\omega, \cdot) \right\|_{L^q(S;X)} \\
&\quad \le \left(\int_S \mathbb{E}_\varepsilon \left\| \sum_{j\in J} \varepsilon_j \lambda_j E_j F(\omega, s) \right\|_X^q d\nu(s) \right)^{1/q} \\
&\quad \lesssim_q \left(\int_S \left[\mathbb{E}_\varepsilon \left\| \sum_{j\in J} \varepsilon_j \lambda_j E_j F(\omega, s) \right\|_X \right]^q d\nu(s) \right)^{1/q} \\
&\quad \le \left(\int_S [M^X_{R,J}(F(\cdot, s))(\omega)]^q \, d\nu(s) \right)^{1/q}.
\end{aligned}$$

Take the supremum over λ, then the $L^q(\Omega)$ norm, and use Fubini and the RMF estimate for X at exponent q. □

5.2 Finite representability

Theorem 5.2 (Locality of RMF)**.** *Suppose there are constants $\lambda, C < \infty$ such that every finite-dimensional subspace $E \subset X$ admits an isomorphism $T_E : E \to Y_E$ into a Banach space Y_E satisfying*

$$\|T_E\| \, \|T_E^{-1}\| \le \lambda, \qquad \rho_p(Y_E) \le C.$$

Then X has RMF_p with constant at most λC. In particular, RMF passes to spaces finitely representable in a fixed RMF space, and to ultraproducts of spaces whose RMF_p constants are uniformly bounded.

Proof. It is enough first to consider a simple X-valued function f and a finite set J. All values of f, and hence all $E_j f$, lie in one finite-dimensional subspace $E \subset X$. After rescaling T_E, assume $\|T_E\| \le 1$ and $\|T_E^{-1}\| \le \lambda$. Since $E_j(T_E f) = T_E(E_j f)$,

$$M_{R,J}^X f \le \|T_E^{-1}\| M_{R,J}^{Y_E}(T_E f) \le \lambda M_{R,J}^{Y_E}(T_E f).$$

Taking L^p norms and using $\rho_p(Y_E) \le C$ gives the result for simple functions. Density and Fatou's lemma give the general case. For ordinary finite representability in a fixed space, apply the theorem with every $\lambda > 1$ and then let $\lambda \downarrow 1$. The ultraproduct assertion follows from the standard fact that each finite-dimensional subspace of an ultraproduct is, up to arbitrarily small distortion, represented in one of the factors. □

5.3 A finite-dimensional dichotomy

Let $\beta_p(X)$ denote the UMD constant of X at exponent p.

Proposition 5.3 (Quantitative reformulation)**.** *Fix $1 < p < \infty$. The assertion that every UMD space has RMF is equivalent to the following quantitative finite-dimensional statement:*

> *For every $B < \infty$ there is $C(B, p) < \infty$ such that every finite-dimensional Banach space E with $\beta_p(E) \le B$ satisfies $\rho_p(E) \le C(B, p)$.*

Consequently, a negative solution can be obtained by finding finite-dimensional spaces E_n with uniformly bounded UMD constants and $\rho_p(E_n) \to \infty$.

Proof. If the quantitative statement holds and X is UMD, every simple X-valued function takes values in a finite-dimensional $E \subset X$ with $\beta_p(E) \le \beta_p(X)$. The uniform estimate on E, followed by density, gives RMF for X.

Conversely, suppose every UMD space has RMF but the quantitative statement fails for some B. Choose finite-dimensional E_n with $\beta_p(E_n) \le B$ and $\rho_p(E_n) \ge n$. Form

$$X = \left(\bigoplus_{n=1}^{\infty} E_n \right)_{\ell^p}.$$

Because $L^p(\Omega; X) = \ell_n^p(L^p(\Omega; E_n))$, the martingale transform inequality holds coordinatewise, so $\beta_p(X) \le B$. Thus X is UMD and, by assumption, has RMF. Each E_n is an isometric closed subspace of X, so Theorem 4.2(i) would give a uniform bound on $\rho_p(E_n)$, a contradiction. □

Proposition 5.3 identifies a concrete counterexample program: one need not guess an infinite exotic space at the outset; it is enough to produce a uniformly UMD sequence of finite-dimensional spaces whose RMF constants diverge.

6 A variation criterion for RMF

This section gives a further sufficient condition. The first ingredient is a deterministic estimate for the R-bound of a finite path.

For a finite set $A \subset X$, define its vector R-bound by

$$\mathcal{R}(A) = \sup \left\{ \mathbb{E}_\varepsilon \left\| \sum_{k=1}^m \varepsilon_k \lambda_k a_k \right\| : m \ge 1,\ a_k \in A,\ \sum_{k=1}^m |\lambda_k|^2 \le 1 \right\}.$$

Repetitions among the a_k are allowed. For an ordered path $x = (x_0, \dots, x_N)$ and $1 \le r < \infty$, set

$$V^r(x) = \sup_{0 \le n_0 < \cdots < n_m \le N} \left(\sum_{i=1}^m \|x_{n_i} - x_{n_{i-1}}\|^r \right)^{1/r}.$$

Lemma 6.1 (Finite-set estimate from type)**.** *If X has Rademacher type $t \in [1,2]$ with type constant $T_t(X)$, and $D \subset X$ has cardinality at most M, then*

$$\mathcal{R}(D) \lesssim_t T_t(X)\, M^{1/t - 1/2} \sup_{d \in D} \|d\|.$$

Proof. Write the distinct elements as $d_1, \dots, d_M$. Given a repeated selection $a_k \in D$ and scalars λ_k, group the indices for which $a_k = d_i$ and put

$$b_i = \left(\sum_{k : a_k = d_i} |\lambda_k|^2 \right)^{1/2}.$$

The grouped scalar sums are independent centered subgaussian random variables with subgaussian parameters b_i. The standard subgaussian–Gaussian comparison, followed by Gaussian type t (equivalent to Rademacher type), gives

$$\mathbb{E} \left\| \sum_k \varepsilon_k \lambda_k a_k \right\| \lesssim_t T_t(X) \left(\sum_{i=1}^M b_i^t \|d_i\|^t \right)^{1/t}.$$

Since $\sum_i b_i^2 = \sum_k |\lambda_k|^2 \le 1$, $\|b\|_{\ell^t} \le M^{1/t-1/2} \|b\|_{\ell^2}$, proving the claim. ☐

Theorem 6.2 (Variation controls the R-bound)**.** *Let X have type $t \in (1,2]$, and let $1 \le r < \infty$ satisfy*

$$r \left(\frac{1}{t} - \frac{1}{2} \right) < 1. \tag{6}$$

Then every finite path $(x_j)_{j=0}^N \subset X$ satisfies

$$\mathcal{R}(\{x_0, \dots, x_N\}) \lesssim_{t,r,T_t(X)} \|x_0\| + V^r(x). \tag{7}$$

For $t = 2$, condition (6) is automatic.

Proof. Write $V = V^r(x)$ and assume $V > 0$. Any δ-separated subset of the range of the path has cardinality at most

$$1 + (V/\delta)^r. \tag{8}$$

Indeed, order its points by their first occurrence along the path. Consecutive selected points are more than δ apart, and their increments form an admissible subsequence in the definition of V^r.

Set $\delta_k = 2^{1-k} V$. Starting with $S_0 = \{x_0\}$, choose for each k a maximal δ_k-separated subset S_k of the path range containing x_0. For a sufficiently large finite K, S_K is the whole distinct range. Since S_{k-1} is a δ_{k-1}-net, choose a parent map $p_k : S_k \to S_{k-1}$ with $\|y - p_k(y)\| \le \delta_{k-1}$. Put

$$D_k = \{ y - p_k(y) : y \in S_k \}.$$

Every point of the path telescopes along its chain of parents, so

$$\{x_0, \dots, x_N\} \subset \{x_0\} + D_1 + \cdots + D_K.$$

The R-bound is subadditive under Minkowski sums. By (8), $|D_k| \le |S_k| \lesssim 2^{kr}$, while $\sup_{d\in D_k} \|d\| \le \delta_{k-1} \lesssim 2^{-k}V$. Hence Lemma 6.1 gives, with $\alpha = 1/t - 1/2$,

$$\mathcal{R}(D_k) \lesssim T_t(X)V\, 2^{-k(1-r\alpha)}.$$

Condition (6) makes this geometric series summable, which yields (7). The case $V = 0$ is immediate. □

The second ingredient is the Pisier–Xu strong variation inequality. A Banach space is called q_0-convexifiable, for $2 \le q_0 < \infty$, if it admits an equivalent uniformly convex norm of power type q_0. Equivalently, it satisfies the corresponding martingale cotype q_0 inequality. Pisier–Xu proved that if X is q_0-convexifiable and $q > q_0$, then for every $1 < p < \infty$ and every finite X-valued martingale M,

$$\|V^q(M)\|_{L^p} \lesssim_{p,q,X} \left\|\sup_n \|M_n\|\right\|_{L^p}. \tag{9}$$

See [7, Theorem 4.2].

Corollary 6.3 (A type–martingale-cotype criterion)**.** *Suppose X has Rademacher type $t \in (1,2]$ and is q_0-convexifiable, where*

$$q_0 < \frac{2t}{2-t}, \tag{10}$$

with the right side interpreted as ∞ when $t = 2$. Then X has RMF.

Proof. Choose q with

$$q_0 < q < \frac{2t}{2-t}.$$

The upper inequality is exactly $q(1/t - 1/2) < 1$. For a finite martingale $M_j = E_j f$, Theorem 6.2 gives pointwise

$$M_R f = \mathcal{R}(\{M_j\}) \lesssim \|M_0\| + V^q(M).$$

Take L^p norms, apply (9), and then Doob's maximal inequality:

$$\|M_R f\|_p \lesssim \|M_0\|_p + \left\|\sup_j \|M_j\|\right\|_p \lesssim_p \|f\|_{L^p(\Omega;X)}.$$

The estimate is uniform in the length of the martingale. □

Remark 6.4. *Corollary 6.3 recovers the type-2 result and covers many spaces whose type and martingale cotype are sufficiently close. It does not cover all UMD spaces. For example, mixed-norm lattices can have type close to 1 and martingale cotype very large, while still having RMF by the lattice argument or by repeated use of Theorem 5.1. Thus the criterion is a genuine sufficient condition, not a characterization.*

7 Why the remaining UMD step is difficult

A useful way to see the obstruction is to fix a finite martingale $M_j = E_j f$. For every *deterministic* $\lambda \in B_{\ell^2}$, Bourgain's vector-valued Stein inequality in a UMD space controls the randomized family of conditional expectations. RMF asks for the pointwise operator norm, so the maximizing coefficients may depend on the terminal sample point:

$$\lambda = \lambda(\omega).$$

Writing $M_j = \sum_{k\le j} d_k$ gives

$$\sum_j \varepsilon_j \lambda_j(\omega) M_j(\omega) = \sum_k \left(\sum_{j\ge k} \varepsilon_j \lambda_j(\omega) \right) d_k(\omega).$$

The coefficient of d_k is not predictable; it can depend on all future martingale differences through $\lambda(\omega)$. The UMD martingale-transform inequality controls predictable multipliers, and this anticipative dependence is precisely what the straightforward proof does not handle.

The closure results suggest a structural route: show that every UMD space lies in the closure of the known RMF classes under subquotients, interpolation, Bochner spaces, and finite representability. No such structural theorem is currently known. A complementary negative route is Proposition 5.3: construct uniformly UMD finite-dimensional spaces with diverging RMF constants. No such sequence was found in this investigation.

8 Consequences for the source Kato theorem

Let $\mathcal{C}$ be the smallest class containing the RMF examples in [3] and closed under:

- closed subspaces and quotients;
- complex interpolation;
- Bochner L^q formation for $1 < q < \infty$;
- the locality operation of Theorem 5.2, whenever the distortion and RMF constants are uniform.

Then every member of $\mathcal{C}$ has RMF. The Banach-valued Kato theorem of [3] therefore applies whenever the UMD space X and its dual X^* belong to this enlarged class.

This extends the supplied packet, but it still does not show that all UMD spaces belong to $\mathcal{C}$.

9 Final assessment

The supplied partial result is mathematically sound after the two repairs given above. The quotient and interpolation claims are stronger than what is explicitly stated in the standard RMF references I located, although the subspace claim is known. The additional Bochner and finite-representability closure theorems, the finite-dimensional dichotomy, and the variation criterion substantially enlarge the partial result.

No full proof of UMD $\Rightarrow$ RMF and no UMD counterexample was obtained. The most concrete next target is the finite-dimensional quantitative problem in Proposition 5.3.

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